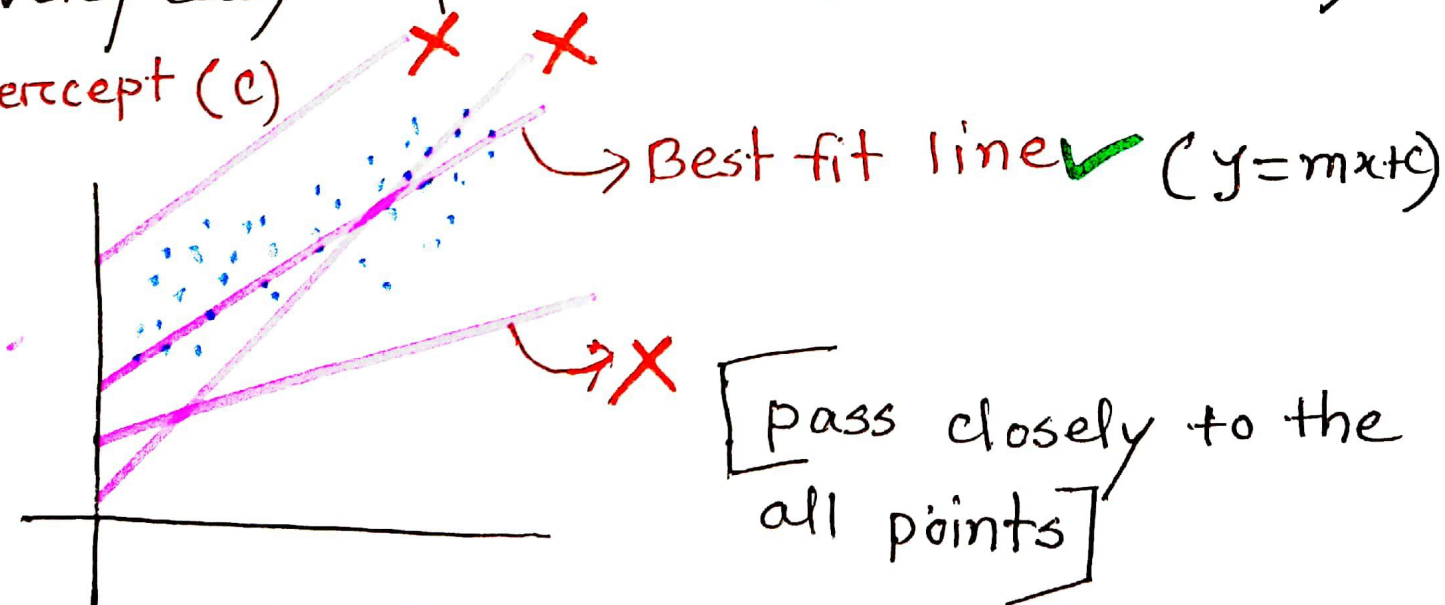
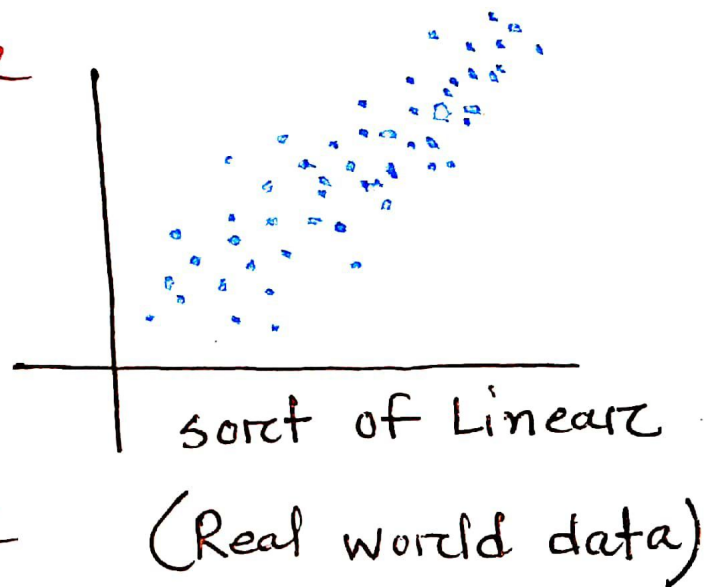
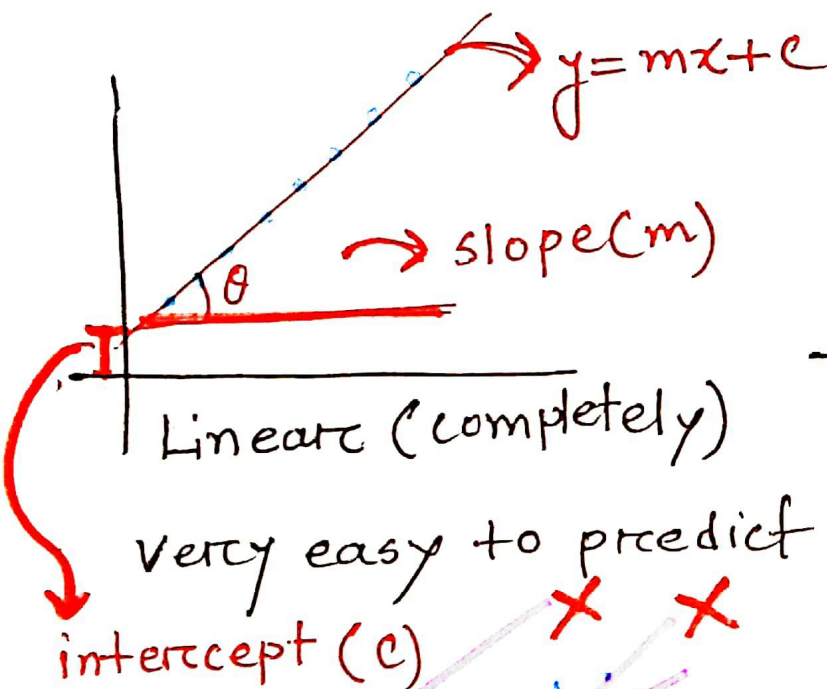
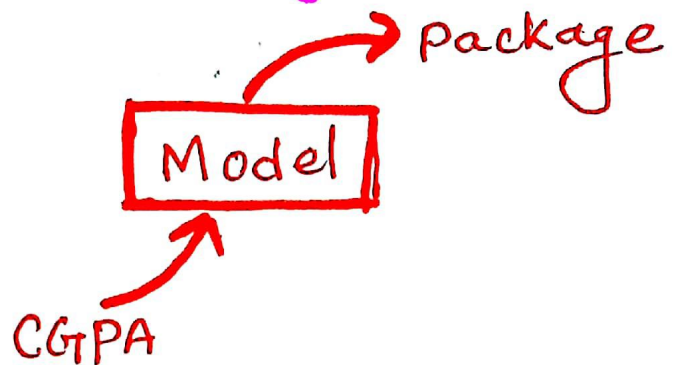


Simple Linear Regression

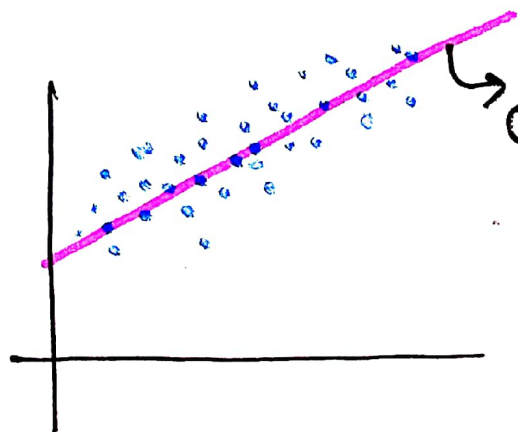
CGPA	Package
7.1	3.5
4.8	1.2
9.8	5.2
8.1	3.9



Himel Sarder

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Find value of m and b



$$y = mx + c$$

(m, c)

direct formula

OLS

closed form solution

Non-closed form solution

approximate based technique

OLS

Gradient Descent

Ordinary Least squares

Not useful for Higher Dimension

Scikit learn

Useful and popular

SGDRegressor

Formula

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

x = Training Example feature values

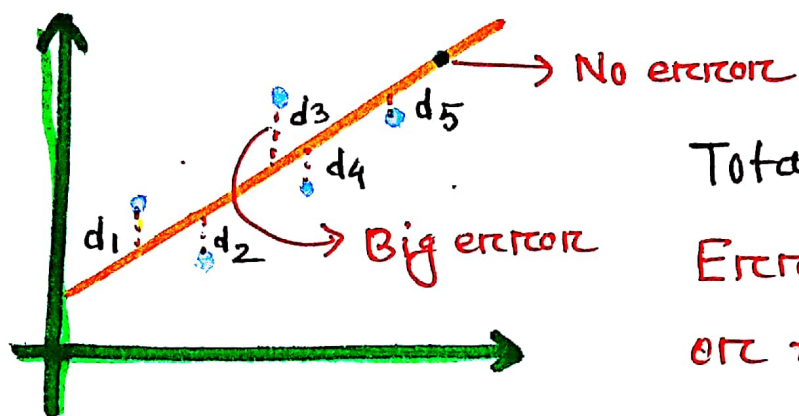
y = Training Example targets

$$b = \bar{y} - m\bar{x}$$

x_i, y_i = i th training examples

$\bar{x}$ = mean of x

$\bar{y}$ = mean of y



$$\text{Total Error} = d_1 + d_2 + d_3 + \dots + d_n$$

Errors can be positive or negative

→ A positive error means the prediction is too low

→ A negative error means the prediction is too high

If we simply summed the errors, positive and negative errors could cancel each other out, giving a misleading total error of zero even when the predictions are inaccurate

So we can

$$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

so why not $E = |d_1| + |d_2| + |d_3| + \dots + |d_n|$

Because it is not differential at origin

$$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

$$E = \sum_{i=1}^n d_i^2 \leftarrow \text{Error function / loss function}$$

Denoted by J

$d_i = (y_i - \hat{y}_i)$ package according to model /
 $\downarrow$ Actual package in data Prediction
 $\hat{y}$ ($\hat{y}$)

Error now for every data

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad E = \text{total error}$$

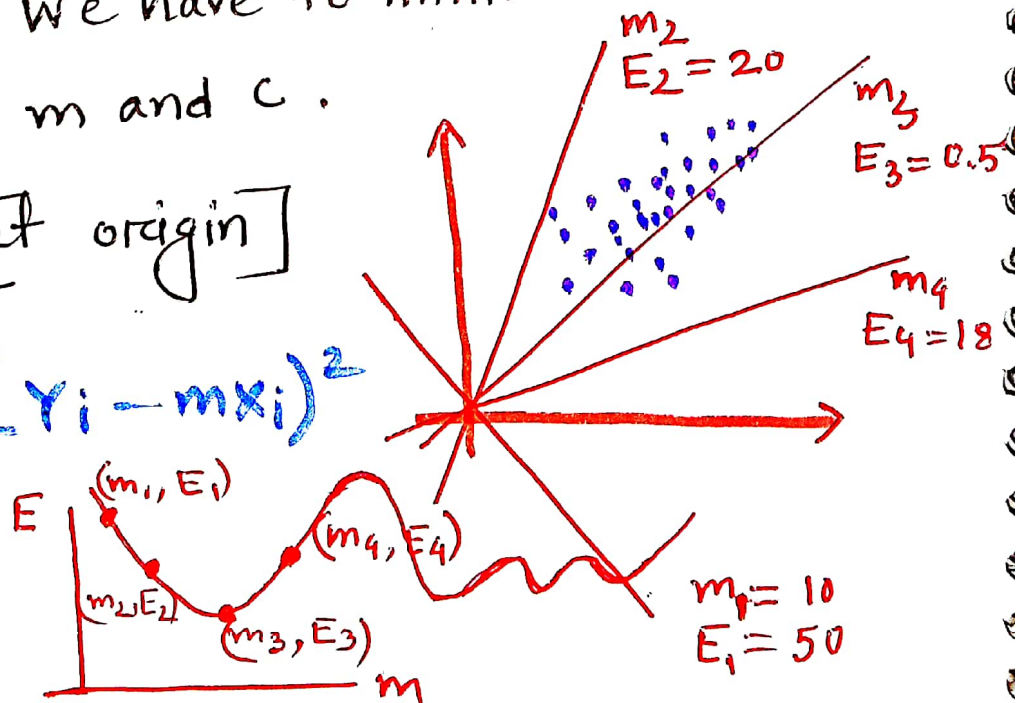
From $\hat{y}_i = mx_i + c$

$$E(m, c) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

We have to minimum it with m and c .

$c = 0$ [at origin]

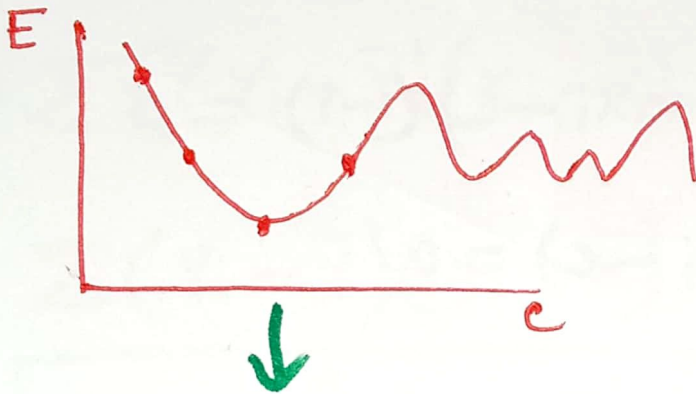
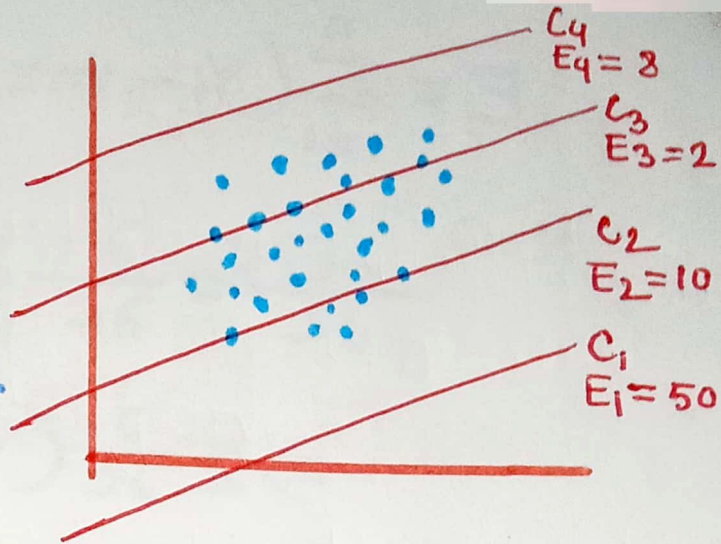
$$E(m) = \sum_{i=1}^n (y_i - mx_i)^2$$



$$m=0$$

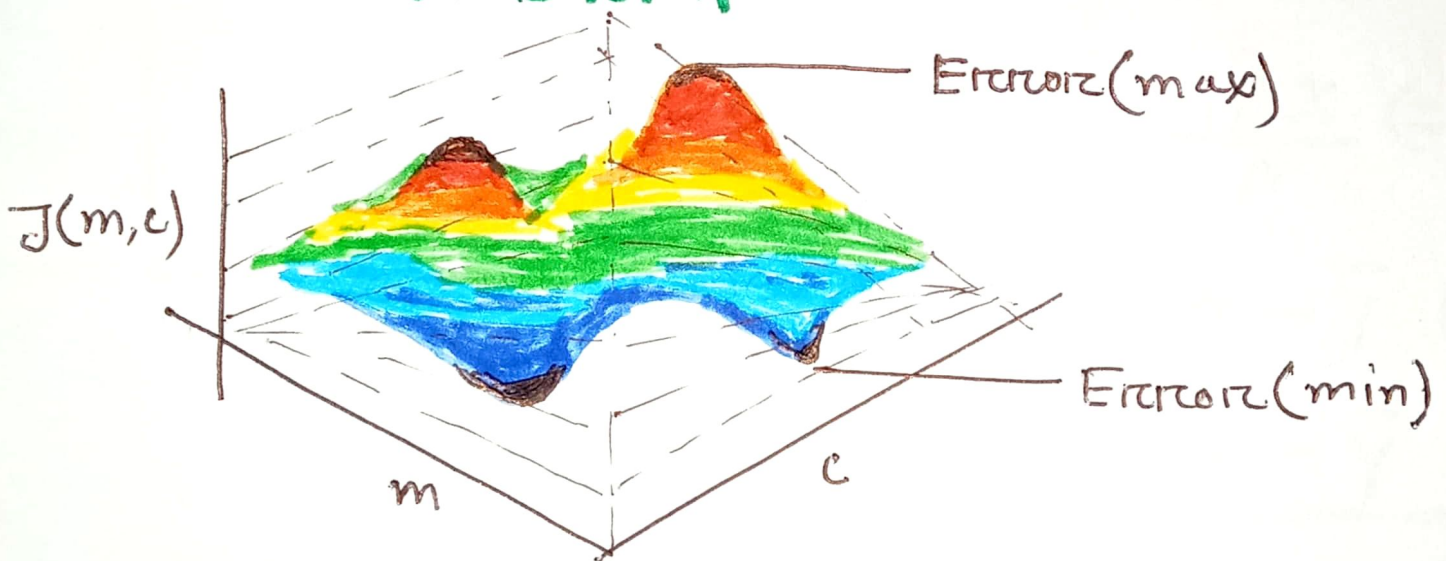
$$E(b) = \sum_{i=1}^n (Y_i - x_i - b)^2$$

$$E(c) = \sum_{i=1}^n (Y_i - x_i - c)^2$$



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For 3-Dimension



Maxima and Minima
 using partial derivatives $\rightarrow$

$$E = \sum_{i=1}^n (Y_i - mx_i - c)^2$$

$$\text{Now, } \frac{\partial E}{\partial c} = \frac{\partial}{\partial c} \sum_{i=1}^n (Y_i - mx_i - c)^2 = 0$$

$$\Rightarrow \sum \frac{\partial}{\partial c} (Y_i - mx_i - c)^2 = 0$$

$$\Rightarrow \sum 2(Y_i - mx_i - c)(-1) = 0$$

$$\Rightarrow \sum (Y_i - mx_i - c) = 0$$

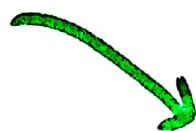
$$\Rightarrow \sum Y_i - \sum mx_i - \sum c = 0$$

$$\Rightarrow \frac{\sum Y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum c}{n} = 0$$

$$\Rightarrow \bar{Y} - m\bar{x} - \frac{nc}{n} = 0$$

$$\bar{Y} - m\bar{x} - c = 0$$

$$\boxed{c = \bar{Y} - m\bar{x}}$$



$$E = \sum (Y_i - mx_i - \bar{Y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (Y_i - mx_i - \bar{Y} + m\bar{x}) = 0$$

$$\Rightarrow \sum 2(Y_i - mx_i - \bar{Y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\Rightarrow \sum -2(Y_i - mx_i - \bar{Y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\Rightarrow \sum (Y_i - mx_i - \bar{Y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\Rightarrow \sum [(Y_i - \bar{Y}) - m(x_i - \bar{x})](x_i - \bar{x}) = 0$$

$$\Rightarrow \sum [(Y_i - \bar{Y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$$

$$\Rightarrow \sum (Y_i - \bar{Y})(x_i - \bar{x}) = m \sum (x_i - \bar{x})^2$$

$$\Rightarrow \boxed{m = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$